Mask-Augmented Multimodal Animal Behaviour Recognition

Does the animal's foreground silhouette *shape* help recognise behaviour, where background segmentation maps did not?

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Abstract

Multimodal models recognise animal behaviour from video, and increasingly from audio and pose. This dissertation asks a narrow, untested question: does adding the animal's *foreground silhouette*—its outline shape, tracked through a clip—improve behaviour recognition over video(+audio) baselines, and is any benefit concentrated in the posture- and contact-defined behaviours where shape should matter most? The motivation is a specific published result: on the MammAlps benchmark, *background/scene* segmentation maps did not help recognition, while audio gave a small lift. Existing mask-based work uses segmentation only to *remove* the background or to crop a region of interest; none treats the silhouette shape itself as a fused, positive input. The approach is deliberately compute-light: every heavy encoder is frozen and its features cached once, so only a small fusion head is trained. The contribution is the answer and its per-behaviour analysis, rather than the model wiring.

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AI Usage Declaration

I acknowledge the use of AI-based tools during the preparation of this dissertation.

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Chapter 1

Introduction

1.1 Motivation

[Draft] Recognising what an animal is doing from camera footage underpins both ecology and neuroscience. Recent benchmarks combine several signals—video, audio, and reference scene-segmentation maps—yet the representation that discards the background entirely and keeps only the animal's shape has not been tested as a positive input.

1.2 Contributions

- A reproducible, frozen-feature **video** and **video+audio** baseline on MammAlps Benchmark I, scored with the authors' official evaluation script.
- A **foreground-silhouette shape** modality (SAM 2 masks $\rightarrow$ shape descriptors) fused via a light, capacity-matched head with modality dropout.
- An **ablation ladder** (video $\rightarrow$ +mask $\rightarrow$ +audio $\rightarrow$ +audio+mask) and a **per-behaviour** analysis focused on posture- and contact-defined classes, plus a background-shift robustness test.

Chapter 2

Background and Related Work

This chapter reviews the work this dissertation builds on and positions its contribution. It covers multimodal behaviour-recognition benchmarks and the role of the background (Sections 2.1–2.2), the use of segmentation and masks in action recognition (Section 2.3), pose-based behaviour analysis (Section 2.4), and modality-fusion under missing inputs (Section 2.5), before summarising the gap this work addresses (Section 2.7).

2.1 Multimodal animal behaviour recognition

Recognising behaviour from camera footage has shifted from single-stream video classifiers to multimodal models. MammAlps [2] is a multimodal, multi-view wildlife benchmark providing video, audio, and reference scene-segmentation maps, with a hierarchical label space of activities and actions across several species and a VideoMAE-based baseline [9]. Two of its findings frame this dissertation. First, audio gives a small but real improvement over video alone. Second, the reference *scene* segmentation maps—a single semantic map of the background per fixed camera viewpoint, essentially static across a clip—did *not* improve over the video or video+audio models. In other words, the benchmark already segments, but it segments the scene rather than the animal, and that scene information does not help the model read behaviour.

2.2 The role of the background

PanAf-FGBG [1] pairs each chimpanzee video with a matching empty-background video and uses SAM 2 [6] masks to *erase* the animal and synthesise backgrounds. Its purpose is to expose how strongly models lean on behaviour-correlated background, and how that reliance harms out-of-distribution generalisation across camera locations. Crucially, the mask there is a tool for *removing* the foreground: the object of study is the background, not the animal's shape.

2.3 Masks and segmentation in action recognition

Mask-guided action recognition has been explored, but typically as a means of background suppression rather than as a shape signal in its own right. MAROON [7], for example, forms background-removed cutouts (via Mask R-CNN) and feeds them as an additional stream, without treating the silhouette shape as a dedicated, fused modality and without pose. The Segment Anything Model 2 (SAM 2) [6] makes per-frame foreground segmentation cheap and promptable—masks can be initialised from animal tracks or boxes and propagated across a clip—which is the enabling tool for extracting a foreground silhouette at scale.

2.4 Pose-based behaviour analysis

Pose-based pipelines encode the animal's skeleton rather than raw pixels. Spatial-temporal graph networks such as ST-GCN [11] model keypoint trajectories, and pose-driven behaviour classifiers such as PoseR [5] are lightweight and effective. Pose is compact, but keypoints capture body *configuration* only sparsely, and they weakly capture two-body *contact*, where the relevant signal is how two bodies overlap and touch. LookAgain [3] reports exactly this asymmetry: added visual context helps some contact behaviours yet not those whose geometry pose already pins down—motivating a dedicated shape stream for contact- and posture-defined behaviours.

2.5 Missing-modality robustness and fusion

For an added modality to be practical it must be droppable at test time. ActionMAE [10] trains with modality dropout so that a multimodal model degrades gracefully when an input is absent. This design directly informs the fusion head used here, allowing the silhouette stream to be omitted at inference without retraining. Where a richer appearance feature is needed, frozen self-supervised encoders such as DINOv2 [4] provide strong masked-crop embeddings, held in reserve should hand-crafted shape descriptors underperform.

2.6 Datasets

MammAlps [2] is the primary dataset: it natively carries video, audio and behaviour labels and supplies the published baseline to reproduce. MABe22 [8] (mouse-triplet subset: clean top-view video and named contact behaviours, but no audio) serves as a controlled second setting in which inter-animal contact geometry can be tested directly, which MammAlps's largely single-animal clips cannot.

2.7 Summary and research gap

If the background does not help behaviour recognition [2] and models over-rely on it [1], the obvious untested question is whether the one representation that discards the background entirely—the animal's foreground silhouette *shape*—adds signal. Unlike scene segmentation, a per-frame foreground silhouette is dynamic: its changing outline is posture, and the overlap of two animals' masks is contact. Treating this silhouette as a fused, positive modality (rather than a background-removal tool or an ROI crop) is unaddressed, and is the gap this dissertation targets.

Chapter 3

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